Extra Dessert #4: Propagation Effects

Transequatorial, long-path, sporadic-E, chordal hop, ground wave, and ordinary/extraordinary waves.

Group E3B · 13 pool questions · Self-study reference · Practice exams at eham.net/exams

Transequatorial Propagation (TEP)

Question	Answer	Q
Where TEP happens	**Between points separated by 2,000-3,000 miles over a path perpendicular to the geomagnetic equator**	E3B01
Maximum TEP range	**About 5,000 miles**	E3B02
Time of day for TEP	**Afternoon or early evening**	E3B03

Long-Path And 160 m

Question	Answer	Q
Best 160 m path	**A path entirely in darkness**	E3B05
Bands where long-path is most frequent	**40 m and 20 m**	E3B06

Ordinary And Extraordinary Waves

Question	Answer	Q
What are O and X waves?	**Independently propagating, elliptically polarized waves created in the ionosphere** (Earth's magnetic field splits a linear signal into two components)	E3B04

Chordal Hop

Question	Answer	Q
What is chordal-hop?	**Successive ionospheric refractions without an intermediate reflection from the ground**	E3B12
Effect of chordal-hop	**Less loss** compared to multi-hop propagation (skipping the ground reflection step)	E3B10

Sporadic-E (Es)

Question	Answer	Q
Time of year	**Around the solstices, especially the summer solstice**	E3B09
Time of day	**Between sunrise and sunset** (daytime)	E3B11

Ground Wave

Question	Answer	Q
How range changes with frequency	**It decreases** (higher frequency = more ground loss = shorter ground-wave range)	E3B08
Polarization supported by ground wave	**Vertical**	E3B13

Takeoff Angle

Question	Answer	Q
Effect of lowering elevation angle on HF skip	**The distance covered by each hop increases**	E3B07

Dessert 4 Question Bank (E3B)

E3B01. Where is transequatorial propagation (TEP) most likely to occur?
A. Between points separated by 2,000 miles to 3,000 miles over a path perpendicular to the geomagnetic equator.
B. Between points located 1,500 miles to 2,000 miles apart on the geomagnetic equator.
C. Between points located at each other's antipode.
D. Through the region where the terminator crosses the geographic equator.

E3B02. What is the approximate maximum range for signals using transequatorial propagation?
A. 1,000 miles.
B. 2,500 miles.
C. 5,000 miles.
D. 7,500 miles.

E3B03. At what time of day is transequatorial propagation most likely to occur?
A. Morning.
B. Noon.
C. Afternoon or early evening.
D. Late at night.

E3B04. What are "extraordinary" and "ordinary" waves?
A. Extraordinary waves exhibit rare long-skip propagation, compared to ordinary waves, which travel shorter distances.
B. Independently propagating, elliptically polarized waves created in the ionosphere.
C. Long-path and short-path waves.
D. Refracted rays and reflected waves.

E3B05. Which of the following paths is most likely to support long-distance propagation on 160 meters?
A. A path entirely in sunlight.
B. Paths at high latitudes.
C. A direct north-south path.
D. A path entirely in darkness.

E3B06. On which of the following amateur bands is long-path propagation most frequent?
A. 160 meters and 80 meters.
B. 40 meters and 20 meters.
C. 10 meters and 6 meters.
D. 6 meters and 2 meters.

E3B07. What effect does lowering a signal's transmitted elevation angle have on ionospheric HF skip propagation?
A. Faraday rotation becomes stronger.
B. The MUF decreases.
C. The distance covered by each hop increases.
D. The critical frequency increases.

E3B08. How does the maximum range of ground-wave propagation change when the signal frequency is increased?
A. It stays the same.
B. It increases.
C. It decreases.
D. It peaks at roughly 8 MHz.

E3B09. At what time of year is sporadic-E propagation most likely to occur?
A. Around the solstices, especially the summer solstice.
B. Around the solstices, especially the winter solstice.
C. Around the equinoxes, especially the spring equinox.
D. Around the equinoxes, especially the fall equinox.

E3B10. What is the effect of chordal-hop propagation?
A. The signal experiences less loss compared to multi-hop propagation, which uses Earth as a reflector.
B. The MUF for chordal-hop propagation is much lower than for normal skip propagation.
C. Atmospheric noise is reduced in the direction of chordal-hop propagation.
D. Signals travel faster along ionospheric chords.

E3B11. At what time of day is sporadic-E propagation most likely to occur?
A. Between midnight and sunrise.
B. Between sunset and midnight.
C. Between sunset and sunrise.
D. Between sunrise and sunset.

E3B12. What is chordal-hop propagation?
A. Propagation away from the great circle bearing between stations.
B. Successive ionospheric refractions without an intermediate reflection from the ground.
C. Propagation across the geomagnetic equator.
D. Signals reflected back toward the transmitting station.

E3B13. What type of polarization is supported by ground-wave propagation?
A. Vertical.
B. Horizontal.
C. Circular.
D. Elliptical.

Answer Key

E3B01: A	E3B02: C	E3B03: C	E3B04: B
E3B05: D	E3B06: B	E3B07: C	E3B08: C
E3B09: A	E3B10: A	E3B11: D	E3B12: B
E3B13: A			